Compilers Assignment-2

1) $S \rightarrow CC$

$C \rightarrow cC$

$C \rightarrow d$

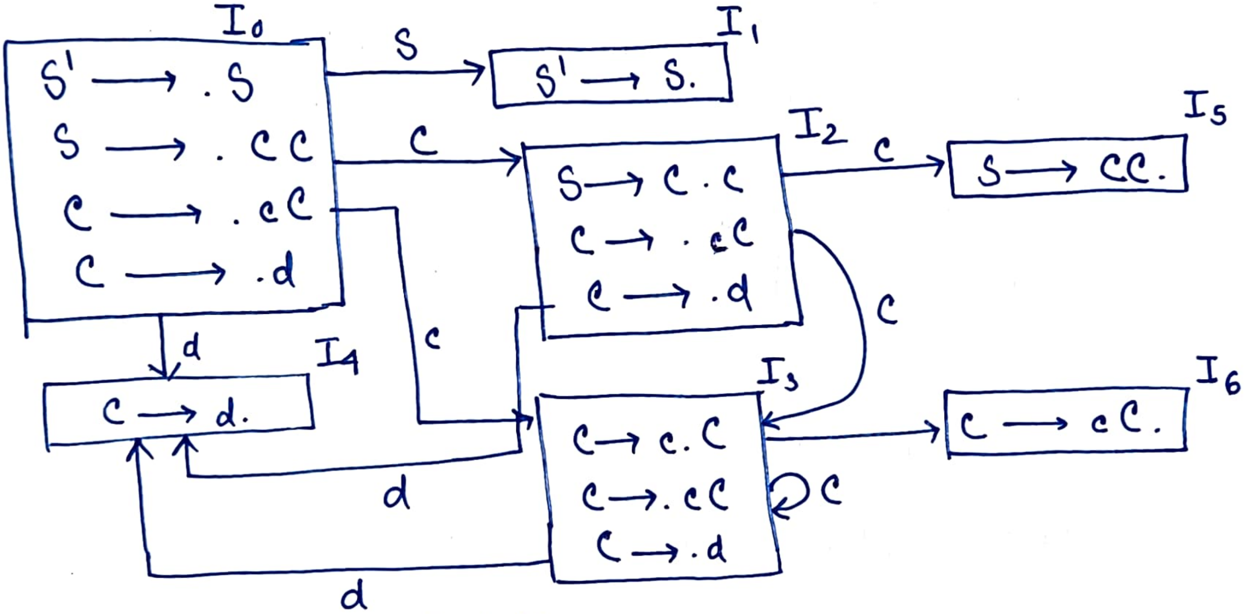
Construct the LR(0) cannonical items. Is the grammar LR(1)? Justify

G': $S' \rightarrow S$

$S \rightarrow CC$

$C \rightarrow cC$

$C \rightarrow d$



	Action			Goto()	
	c	d	\$	S	C
0	S_3	S_4		1	2
1			Accept		
2	S_3	S_4			5
3	S_3	S_4			6
4	r_3	r_3	r_3		
5	r_1	r	r_1		
6	r_2	r_2	r_2		

∴ As there is no conflict, So, the grammar is LR(0)

∴ Every LR(0) grammar is LR(1)

∴ G is LR(1)

Ans

2) construct the SLR Parsing table

	Action			Goto	
	c	d	$	S	C
0	S_3	S_4		1	2
1			Accept		
2	S_3	S_4			5
3	S_3	S_4			6
4	r_3	r_3	r_3		
5			r_1		
6	r_2	r_2	r_2		

$r_3 \Rightarrow C \rightarrow d$

$F_o(C) = \{c, d, \$\}$

$r_2 \Rightarrow C \rightarrow cC$

$F_o(C) = \{c, d, \$\}$

$r_1 \Rightarrow S \rightarrow CC$

$F_o(S) = \{\$\}$

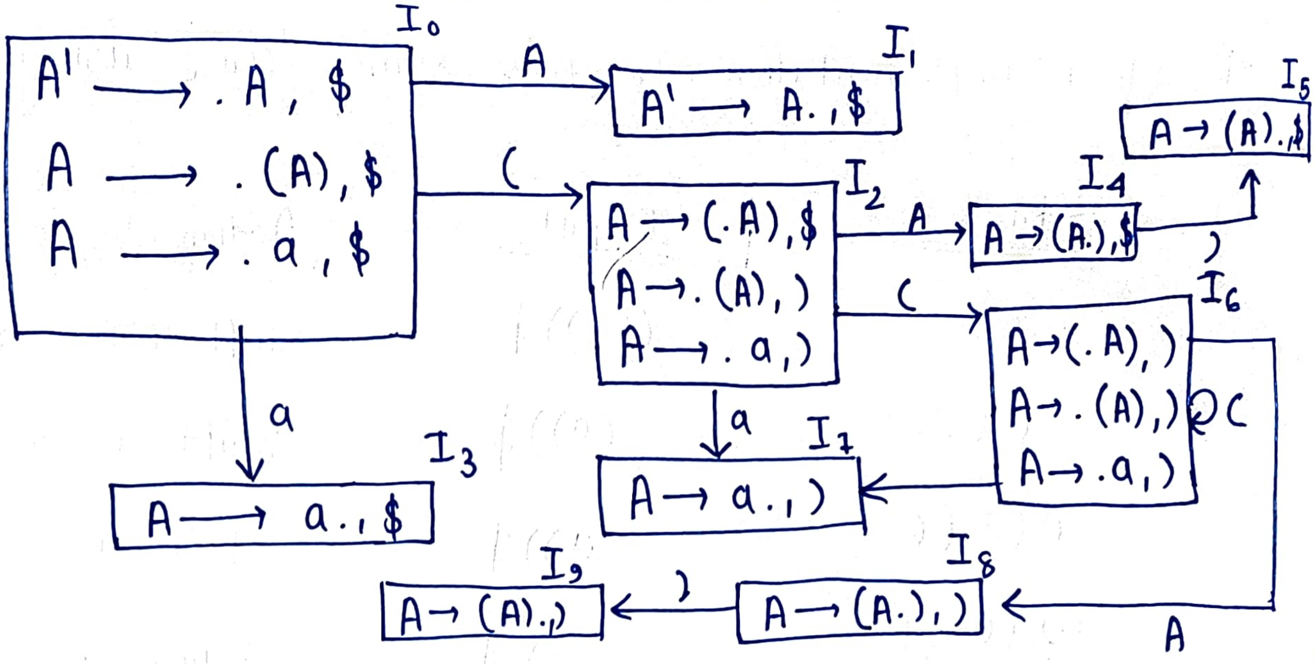
3) $G: A \longrightarrow (A) \mid a$

(i) Construct LR(1) cannonical items

$G': A' \longrightarrow A$

$A \longrightarrow (A)$

$A \longrightarrow a$



(ii) Construct the CLR(1) Parsing table

	Action				Goto()
	(	)	a	$	A
0	S_2		S_3		1
1				Accept	
2	S_6		S_7		4
3				r_2	
4		S_5			
5				r_1	
6	S_6		S_7		8
7		r_2			
8		S_9			
9		r_1			

4) Stack implementation to parse string '((a))' using CLR(1)

Stack	Input	Action
0	((a))$	Shift ($\rightarrow$ S2
0(2	(a))$	Shift ($\rightarrow$ S6
0(2(6	a))$	Shift a $\rightarrow$ S7
0(2(6a7	))$	Reduce A $\rightarrow$ a
0(2(6A8	))$	Shift) $\rightarrow$ S9
0(2(6A8)9	)$	Reduce A $\rightarrow$ (A)
0(2A4	)$	Shift) $\rightarrow$ S5
0(2A4)5	$	Reduce A $\rightarrow$ (A)
0A1	$	Accept

5) Construct the LALR(1) Parsing table

States having same items & diff. lookahead symbols —

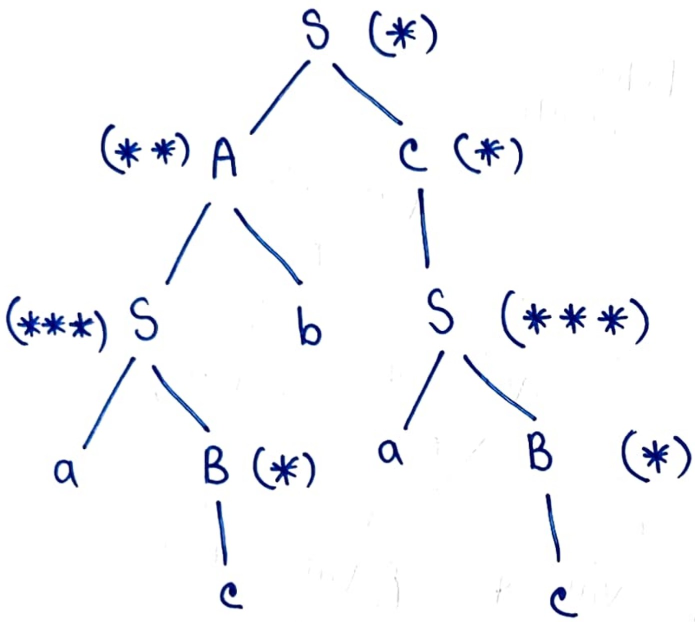
$\{(I_2, I_6), (I_3, I_7), (I_4, I_8), (I_5, I_9)\}$

	Action				Goto
	(	)	a	$	A
0	S_{26}		S_{37}		1
1				Accept	
26	S_{26}		S_{37}		48
37		r_2		r_2	
48		S_{59}			
59		r_1		r_1	

6) $S \rightarrow AC$ {Print *}
$S \rightarrow aB$ {Print ***}
$A \rightarrow Sb$ {Print **}
$B \rightarrow c$ {Print *}
$C \rightarrow S$ {Print *}

i/p: acbac

<u>Parse Tree</u>



7) calculate the total no. of Star (*) Printed for i/p: acbac

o/p: *
*
* * *
* * *
* *
*
*

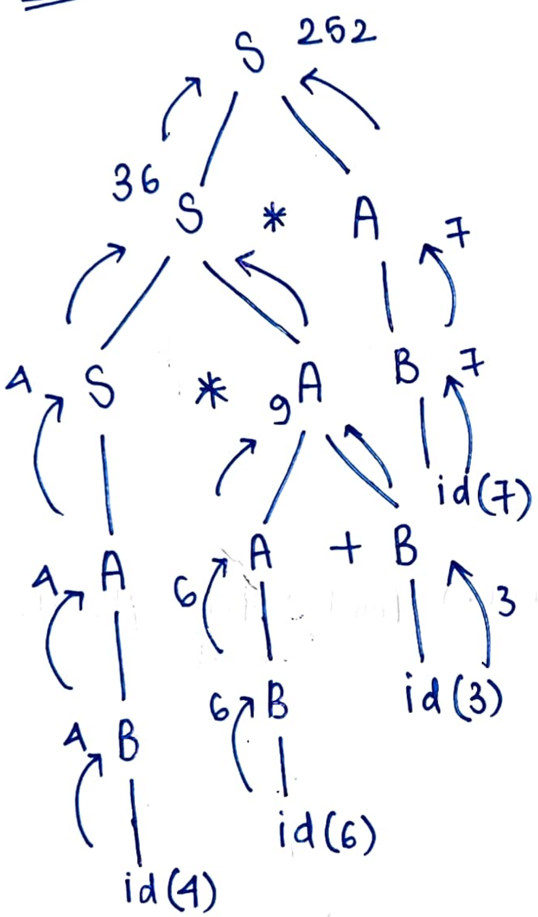
∴ Total no. of (*) = 12 <u>Ans</u>

8)

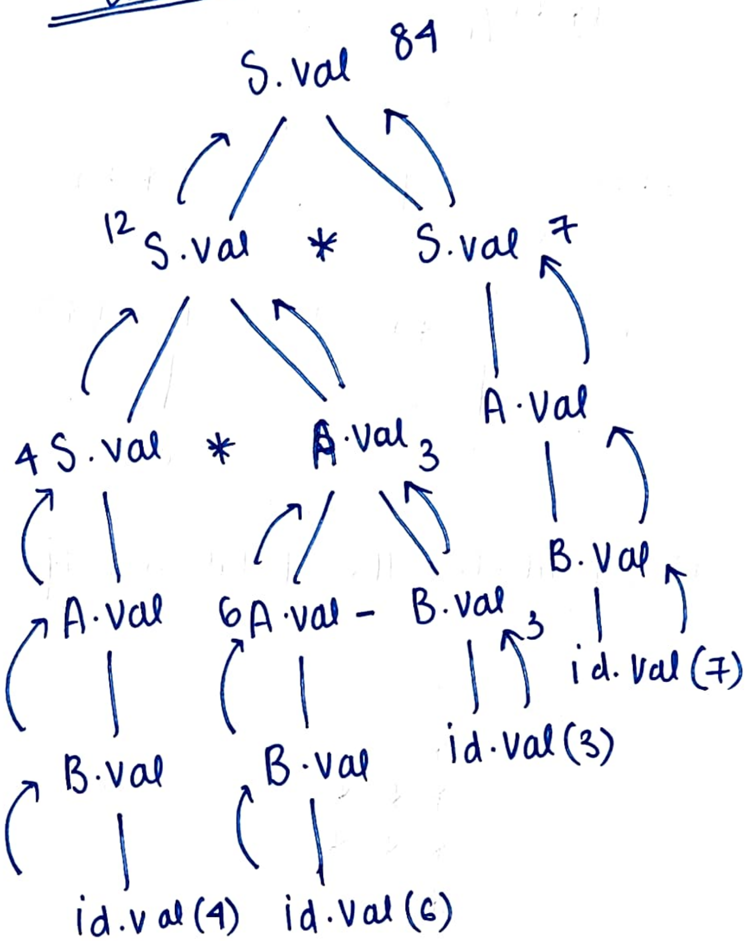
$S \longrightarrow S * A$	$\{S.val = S.val \times A.val\}$
$S \longrightarrow A$	$\{S.val = A.val\}$
$A \longrightarrow A + B$	$\{A.val = A.val - B.val\}$
$B \longrightarrow (S)$	$\{B.val = 2\}$
$A \longrightarrow B$	$\{A.val = B.val\}$
$B \longrightarrow id$	$\{B.val = id.val\}$

i/p: $4 * 6 + 3 * 7$

Parse Tree



Dependency Graph



Answer from parse tree evaluation = 252

∴ Answer from Dependency graph evaluation = 84

Ans

Q) Write the three address code & draw the DAG for expression : $a = (-c * b) + (-c * d)$

Three address code —

$t_1 = -c$

$t_2 = t_1 * b$

$t_4 = t_1 * d$

$t_5 = t_2 + t_4$

$a = t_5$

Optimized —

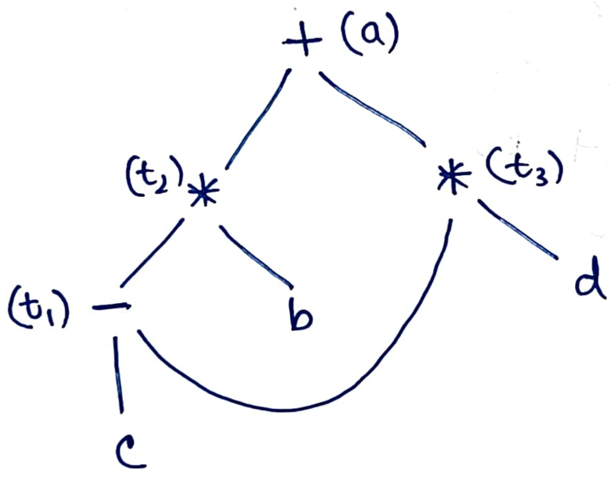
$t_1 = -c$

$t_2 = t_1 * b$

$t_3 = t_1 * d$

$a = t_2 + t_3$

DAG



10) write the assembly lang. code for the optimal intermediate code

Optimal code

$t_1 = -c$

$t_2 = t_1 * b$

$t_3 = t_1 * d$

$a = t_2 + t_3$

Assembly code

```
MOV  R1, c
NEG  R1
MOV  R2, b
MOV  R3, d
MUL  R4, R1, R2
MUL  R5, R1, R3
ADD  R6, R4, R5
MOV  a, R6
```